How Plants Grow

Reading & Seed Observation Workbook

FROM A BEAN SEED TO A YOUNG PLANT



READ CLOSELY • OBSERVE HONESTLY • EXPLAIN WITH EVIDENCE

Three science readings • Seed investigation • Teacher answer key

GRADES 3–4

US LETTER • PRINTABLE PDF

DESIGN PROTOTYPE • Science and commercial rights require human review

Reading 1: From Seed to Seedling

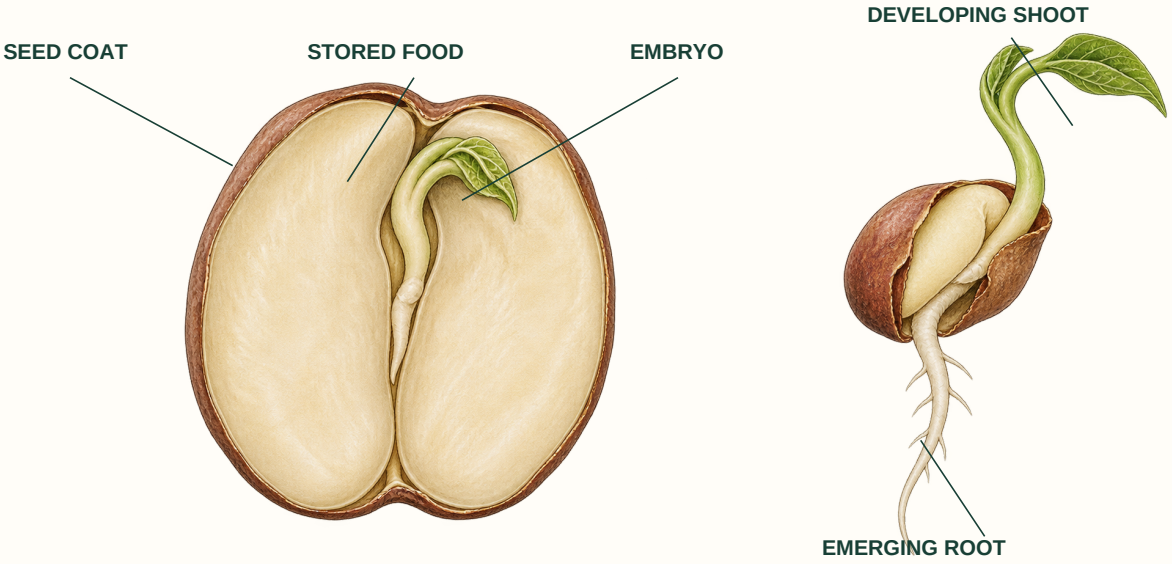
Describe what a bean seed needs to start growing.

NAME _____ DATE _____

A dry bean seed holds an embryo (a young plant) and stored food inside its seed coat. The coat protects the parts inside, although it cannot protect a seed from every danger. A seed begins to germinate when it takes in water under suitable conditions. It also needs air and a suitable temperature. A seed does not have to wait for spring rain if someone provides those conditions indoors.

As the bean takes in water, it swells. The first root grows downward and absorbs water. A shoot then grows upward. The young plant uses food stored in the seed while its first leaves develop. Later, green leaves can make sugars using light energy. This change from seed to seedling takes time, and different seeds may grow at different rates.

LOOK CLOSELY / A BEAN SEED AND A GERMINATING BEAN



WORD HELP

Germination
the beginning of growth from a seed

Seedling
a young plant

Embryo
the young plant inside a seed